

## COSC 39: Theory of Computation

**Problem 1.** Describe a Turing machine that achieves the following task. For this worksheet and this worksheet alone, high-level pseudocodes are not allowed unless the instructions can directly be mapped into transition functions between states in a Turing machine, and the programs are well-commented.

1. Decide if the input is in the language  $\{ww^R : w \in \Sigma_\star\}$ . (These are even-length palindromes.)
2. Given an input string  $0^m1^n$  where  $n$  and  $m$  are positive integers, decide if  $n$  divides  $m$ , and if so, leave  $\#^{m/n}$  on the tape to the right of the input.

*Solution.*

- 1.